Denis Jacob Machado

- **Bioinformaticist**, **zoologist**, and **marine biologist** with a strong background in **parasitology**, **virology**, **biostatistics**, **data science**, **computational biology**, and **phylogenetics**.
- Peer-reviewed publications on **comparative genomics**, and **zoonosis** of flaviviruses and coronaviruses, among other topics.
- 15+ years of experience in **molecular biology** labs, developing and optimizing **high-throughput sequencing** protocols (e.g., Illumina, PacBio, Hi-C, GWAS, and RNA-Seq).
- 8+ years of experience in **machine learning** (e.g., PyTorch, TensorFlow), **big data analyses**, **mathematical modeling**, **high-throughput data processing**, and **data mining algorithms** to solve practical problems in emerging infectious diseases research.
- Concepts used in my current and past projects include Bash, Python, R, Flask, SQL, AWS, Git, Shiny, Tableau, Jupyter, AlphaFold2, RoseTTAFold, and HADDOCK.

Work Experience

Aug. 2022–Current **Assistant Professor**, *University of North Carolina at Charlotte*, Charlotte–NC, USA

- Two main lines of research:
  - Investigate new pathogens' emergence, evolution, and spread, focusing on preventing and treating infectious diseases.
  - Create computational and molecular solutions to make data from biorepositories more readily available to biomedical research.
- For more information, please visit phyloinformatics.com.

2019–2022 **Postdoctoral Researcher**, *University of North Carolina at Charlotte*, Charlotte–NC, USA

- I studied the origins, evolution, and zoonotic events of coronaviruses. To that end, I created the first programs for gene prediction and annotation of *Orthocoronavirinae* (including SARS-COV-2) and *Flaviviridae* (including hepatitis C, yellow fever, dengue, and Zika virus) and deployed the pipelines as a web application. I also employed big data analysis to categorize different variants of SARS-CoV-2 and applied deep learning techniques to produce highly accurate structure predictions of their proteins.
- I use artificial intelligence to predict how structural changes in the receptor-binding domain of the spike protein of different variants of SARS-CoV-2 may reduce antibody interaction without completely evading existing neutralizing antibodies (and therefore current vaccines).
- I performed comparative genomic analyses of highly regenerative echinoderms to develop new models for the study of tissue regeneration in humans.

2018–2019 **Research Collaborator**, *University of São Paulo*, São Paulo–SP, Brazil

- I built and administered the first computer clusters at the Museum of Zoology of the University of São Paulo (MUZUSP), providing training and technology to bridge the gap between animal research and applied bioinformatics with potential biomedical significance.
- I participated in the planning, funding acquisition, and implementation of the first laboratory to sequence historical DNA (e.g., from degraded museum samples) at the Department of Zoology of the University of São Paulo (USP).
- I developed new genome skimming techniques to retrieve genomic data of non-model organisms (mainly parasites and mosquito vectors) when samples were small or degraded. I also created the necessary software to facilitate the analysis of organelle genomes, including new indices that check for the completeness of circular genomes or misalignments in other types of contigs and scaffolds.

2018–2019 **Invited Lecturer**, *University of São Paulo*, São Paulo–SP, Brazil

- I created and taught the first graduate-level courses in computer programming and bioinformatics for biologists at USP's Graduate Program in Zoology, which is ranked first in animal research worldwide.

Jul–Aug, 2017 **Invited Lecturer**, *University of Magdalena*, Santa Marta D.T.C.H., Colombia

- I designed and taught the first graduate-level course in computational biology at the University of Magdalena. Classes focused on the evolution and epidemiology of flaviviruses.

Education

1/2019–8/22 **Postdoctoral Researcher**, *University of North Carolina at Charlotte (UNC Charlotte)*, Charlotte–NC, U.S.A.
Keywords: genomics, transcriptomics, phylogenomics, regeneration, virus

2/2013–4/18 **Ph.D. in Bioinformatics**, *University of São Paulo (USP)*, São Paulo–SP, Brazil
Dissertation: "Comparative genomics and the evolution of amphibian chemical defense." DOI: 10.11606/T.95.2018.tde-20012023-095836
Keywords: high-performance computing, next-generation sequencing, and comparative genomics

2/2010–7/12 **M.Sc. in Zoology**, *University of São Paulo (USP)*, São Paulo–SP, Brazil
Thesis: "*Acanthobothrium* Blanchard, 1848: diversidade filogenética das linhagens parasitas de arraias de água doce."
Keywords: invertebrate zoology, host-parasite coevolution, molecular biology, and phylogenetics

2/2006–12/9 **B.Sc. in Biological Sciences**, *São Paulo State University (UNESP)*, São Vicente–SP, Brazil
Qualification: Marine Biology

Honors & Awards

2026–2028 **Council Member**, *The Willi Hennig Society*, cladistics.org

2025 **Eponym: *Anthocephalum machadoi***, *New tapeworm species (Rhinebothriidea: Anthocephalidae)*, *Zootaxa*
• Sakai, Marques, and Trevisan named this species "for his contribution to the systematics of cestodes through his expertise in bioinformatics."
• DOI: 10.11646/zootaxa.5584.2.1

2024 **Fellow**, *The Willi Hennig Society*, cladistics.org
• Nominated at the XLI Annual Meeting of the Willi Hennig Society, Mexico City, July 2024

2024 **Safety Champion Award**, *UNC Charlotte, Environmental Health & Safety (EHS)*, Charlotte, NC, U.S.A.
• Awarded during the 2024 Safe and Sound Week for outstanding commitment to safety and health practices in the workplace.

2023 **Top Cited Article (2021-2022)**, *Cladistics, Wiley*, Online
• Awarded by Cladistics for the paper: "Fundamental evolution of all *Orthocoronavirinae* including three deadly lineages descendant from Chiroptera-hosted coronaviruses: SARS-CoV, MERS-CoV, and SARS-CoV-2."
• Published in Volume 37, Issue 5, 2021.
• Link: `https://onlinelibrary.wiley.com/doi/full/10.1111/cla.12454`

Dec. 2023 **Green Dot Trainer of the Month**, *The Green Dot Program, UNC Charlotte*, Charlotte, NC, U.S.A.
• Recognized as "Green Dot Trainer of the Month" in The Green Dot Newsletter (December 2023, Vol. 3).
• Acknowledged for outstanding contributions to "violence prevention training and campus safety awareness."
• Green Dot is a bystander intervention program focused on reducing power-based personal violence through education and community engagement.

2016 **Grand Stars in Human Health**, *Grand Challenges Canada*, Canada
• $461,250 for the development of a "Larvicide Automatic Dispenser" (LAD) to be used in the population control of mosquito vectors of flaviviruses including dengue and Zika virus

Apr. 2016 **Most Implementable Solution, sponsored by GE Foundation**, *Zika Innovation Hack-a-thon*, Boston–MA, USA
• Proposal of a "Larvicide Automatic Dispenser" (LAD) device to be used in the population control of mosquito vectors of flaviviruses including dengue and Zika virus

2016–2017 **Research Internship Abroad**, *São Paulo Research Foundation (FAPESP)*, São Paulo–SP, Brazil
• Project: "Whole-genome sequence of the Eastern spadefoot toad, *Scaphiopus holbrookii* (Amphibia: Anura: Scaphiopodidade) and of the Maldonado redbelly toad, *Melanophryniscus moreirae* (Amphibia: Anura: Bufonidae)"
• File number: 2015/18654-2

2013 **The Willi Hennig Award**, *The Willi Hennig Society*, Rostock, Germany

— **Kurt Milton Pickett Award**, *The Willi Hennig Society*, Rostock, Germany

2013–2018 **Doctorate Scholarship Award**, *São Paulo Research Foundation (FAPESP)*, São Paulo–SP, Brazil
• File number: 2013/05958-8

2010–2012 **Master Scholarship Award**, *São Paulo Research Foundation (FAPESP)*, São Paulo–SP, Brazil
• File number: 2009/13561-5

2009 **Scientific Initiation Award**, *São Paulo Research Foundation (FAPESP)*, São Paulo–SP, Brazil
• File number: 2009/00886-3

Selected Publications

You can retrieve my complete list of publications from ORCiD (0000-0001-9858-4515) or Google Scholar (scholar.google.com/citations?user=TK6Ms80AAAAJ). In October 2026, Google Scholar listed 826 citations, an h-index of 13, and an i10-index of 14.

2026

Scorsim Omura, G. Y., Furr, D., & **Jacob Machado, D.** (2026). Eight quick tips for requesting materials from natural history collections. *PLOS Computational Biology*, **22(9)**, e1014753. DOI: 10.1371/journal.pcbi.1014753.

Alves, P. V., da Silva, R. J., Janies, D., Taylor, W., Harris, A., New, G., & **Jacob Machado, D.** (2026). Mining natural history collections for molecular treasures: mitogenomes and nuclear ribosomal transcription units of proteocephalid tapeworms (Onchoproteocephalidea) via genome skimming. *International Journal for Parasitology*, **56(7)**, 104762. DOI: 10.1016/j.ijpara.2025.12.007.

Marne, O., & **Jacob Machado, D.** (2026). Advancing FAIR phylogenetics for health threats: a systematic review of SARS-CoV-2 research and guidelines for future outbreaks. *Cladistics*. DOI: 10.1111/cla.70050.

Rodrigues, F. M., Prieto-Oliveira, P., Zukurov, J. P., Alkmim, W. T., Soane, M. M., Camargo, M., Sanabani, S. S., Kallas, E. G., Sucupira, M. C., Diaz, R. S., **Jacob Machado, D.**, & Janini, L. M. (2026). Multiple infections, recombination, and hypermutation during a 12-month prospective study of five HIV-1 infected individuals. *Microbiology Research*, **17(2)**, 30. DOI: 10.3390/microbiolres17020030. Corresponding author.

2025

Márquez, R., **Jacob Machado, D.**, Nouri, R., Gendreau, K. L., Janies, D., Saporito, R. A., Kronforst, M. R., & Grant, T. (2025). A draft genome assembly for the dart-poison frog *Phyllobates terribilis*. *Gigabyte*, 2025, gigabyte157. DOI: 10.46471/gigabyte.157. Co-first and corresponding author.

Alves, P. V., da Silva, R. J., Scholz, T., de Chambrier, A., Luque, J. L., Duchenko, A., Janies, D., & **Jacob Machado, D.** (2025). Machine learning models accurately predict clades of proteocephalidean tapeworms (Onchoproteocephalidea) based on host and biogeographical data. *Cladistics*, **41(3)**, 264–282. DOI: 10.1111/cla.12610.

2024

Nouri, R., Mashanov, V., Harris, A., New, G., Taylor, W., Janies, D., Reid, R. W., & **Jacob Machado, D.** (2024). Unveiling putative modulators of mutable collagenous tissue in the brittle star *Ophiomastix wendtii*: an RNA-Seq analysis. *BMC Genomics*, 25(1), 1013. DOI: 10.1186/s12864-024-10926-7.

Costa, A. L. D., Prieto-Oliveira, P., Duarte-Barbosa, M., Andreata-Santos, R., Peter, C. M., Prolo de Brito, T., ... & **Jacob Machado, D.** (2024). The Relationship between HERV, Interleukin, and Transcription Factor Expression in ZIKV Infected versus Uninfected Trophoblastic Cells. *Cells*, 13(17), 1491. DOI: 10.3390/cells13171491. Joint last author.

Yasa, S., Guirales-Medrano, S., **Jacob Machado, D.**, Ford, C. T., & Janies, D. (2024). Predicting antibody and ACE2 affinity for SARS-CoV-2 BA. 2.86 and JN. 1 with *in silico* protein modeling and docking. *Frontiers in Virology*, 4, 1419276. DOI: 10.3389/fviro.2024.1419276.

2023

Jacob Machado, D., Jiménez-Ferbans, L., Ocaña, K., and de Bernardi Schneider, A. (2023). Editorial: Phylogenetics in the One Health context. *Frontiers in Microbiology*, 1307616. DOI: 10.3389/fmicb.2023.1307616.

Ford, C. T., Yasa, S., **Jacob Machado, D.**, White III, R. A., and Janies, D. A. (2023). Predicting changes in neutralizing antibody activity for SARS-CoV-2 XBB. 1.5 using in silico protein modeling. *Frontiers in Virology*, **3**, 1172027. DOI: 10.3389/fviro.2023.1172027.

Mashanov, V. S., Ademiluyi, S., **Jacob Machado, D.**, Reid, R., and Janies, D. (2023). Echinoderm radial glia in adult cell renewal, indeterminate growth, and regeneration. *Frontiers in Neural Circuits*, **17**, 1258370. DOI: 10.3389/fncir.2023.1258370.

2022

Jacob Machado D., Marques F.P.L., Jiménez-Ferbans L., and Grant T. (2022) An empirical test of the relationship between the bootstrap and likelihood ratio support in maximum likelihood phylogenetic analysis. *Cladistics*, **38(3)**: 392–401. DOI: 10.1111/cla.12496.

Mittal V., Reid R. W., **Jacob Machado D.**, Mashanov V., and Janies D. A. (2022). EchinoDB: an update to the web-based application for genomic and transcriptomic data on echinoderms. *BMC Genomic Data*, **23(1)**: 75. DOI: 10.1186/s12863-022-01090-6.

Mashanov V., **Jacob Machado D.**, Reid R., Brouwer C., Kofsky J., and Janies D.A. (2022) Twinkle twinkle brittle star: the draft genome of *Ophioderma brevispinum* (Echinodermata: Ophiuroidea) as a resource for regeneration research. *BMC Genomics*, **23**: 574. DOI: 10.1186/s12864-022-08750-y.

Mashanov V., Whaley L., Davis K., Heinzeller T., **Jacob Machado D.**, Reid R.W., Kofsky J., and Janies D.A. (2022) A subterminal growth zone at arm tip likely underlies life-long indeterminate growth in brittle stars. *Frontiers in Zoology*, **19**: 15. DOI: 10.1186/s12983-022-00461-0.

Ford C.T., **Jacob Machado D.**, and Janies D.A. (2022) Predictions of the SARS-CoV-2 Omicron variant (B.1.1.529) spike protein receptor-binding domain structure and neutralizing antibody interactions. *Frontiers in Virology*, **2**: 830202. DOI: 10.3389/fviro.2022.830202.

Jacob Machado D., White III R.A., Kofsky J., and Janies D.A. (2022) Fundamentos de la epidemiología genómica, lecciones aprendidas de la enfermedad por coronavirus (COVID-19) y nuevas direcciones. *Revista Magna Scientia*, Unidad Central del Valle del Cauca. Spanish version of the 2021 article in *Antimicrobial Stewardship & Healthcare Epidemiology*.

2021

Jacob Machado D., White III R.A., Kofsky J., and Janies D.A. (2021) Fundamentals of genomic epidemiology, lessons learned from the coronavirus disease 2019 (COVID-19) pandemic, and new directions. *Antimicrobial Stewardship & Healthcare Epidemiology*, **1**: E60. DOI: 10.1017/ash.2021.222.

Jacob Machado D., Scott R., Guirales S., and Janies D.A. (2021) Fundamental evolution of all *Orthocoronavirinae* including three deadly lineages descendent from Chiroptera-hosted coronaviruses: SARS-CoV, MERS-CoV and SARS-CoV-2. *Cladistics*, **37(5)**: 461–488. DOI: 10.1111/cla.12454.

Jacob Machado D., Castroviejo-Fisher S., and Grant T. (2021) Evidence of absence treated as absence of evidence: the effects of variation in the number and distribution of gaps treated as missing data on the results of standard maximum likelihood analysis. *Molecular Phylogenetics and Evolution*, **154**: 106966. DOI: 10.1016/j.ympev.2020.106966.

Orrico V.G.D., Grant T., Faivovich J., Rivera-Correa M., Rada, M.A., Lyra M.L., Cassini C.S., Valdujo P.H., Schargel W.E., **Jacob Machado D.**, Wheeler W.C., Barrio-Amorós C., Loebmann D., Moravec J., Zina J., Solé M., Sturaro M.J., Peloso P.L.V., Suarez P., and Haddad C.F.B. (2021). The phylogeny of Dendropsophini (Anura: Hylidae: Hylinae). *Cladistics*, **37(1)**: 73–105. DOI: 10.1111/cla.12429.

Trevisan B., **Jacob Machado D.**, Lahr D.J.G., Marques F.P.L.M. (2021) Comparative characterization of mitogenomes from five orders of cestodes (Eucestoda: Tapeworms). *Frontiers in Genetics*, **12**: 788871. DOI: 10.3389/fgene.2021.788871.

2020

de Bernardi Schneider A., **Jacob Machado D.**, Guirales S., and Janies D.A. (2020) FLAVi: an enhanced annotator for viral genomes of *Flaviviridae*. *Viruses*, **12(8)**: 892. DOI: 10.3390/v12080892. Equal contribution with the first author. Corresponding author.

Mashanov V., Akiona J., Khoury M., Ferrier J., Reid R., **Jacob Machado D.**, Zueva O., and Janies D. (2020) Active Notch signaling is required for arm regeneration in a brittle star. *PLoS ONE*, **15(5)**: e0232981. DOI: 10.1371/journal.pone.0232981

Lin J.-P., Tsai M., Kroh A., Trautman A., **Jacob Machado D.**, Chang L.-Y., Reid R., Lin K.-T., Bronstein O., Lee S.-J., and Janies D. (2020) The first complete mitochondrial genome of the sand dollar *Sinaechinocyamus mai* (Echinoidea: Clypeasteroida). *Genomics*, **112(2)**: 1686-1693. DOI: 10.1016/j.ygeno.2019.10.007.

2019

Trevisan B., Alcantara D.M.C., **Jacob Machado D.**, Marques F.P.L. & Lahr D.J.G. (2019) Genome Skimming is a low-cost and robust strategy to assemble complete mitochondrial genomes from ethanol preserved specimens in biodiversity studies. *PeerJ*, **7**: e7543. DOI: 10.7717/peerj.7543.

2018

Jacob Machado D., Janies D., Brouwer C. & Grant T. (2018) A new strategy to infer circularity applied to four new complete frog mitogenomes. *Ecology and Evolution*, **8(8)**: 4011–4018. DOI: 10.1002/ece3.3918.

2017

Grant T., Rada M.A., Anganoy-Criollo M., Batista A., Dias P.H., Jeckel A.M., **Jacob Machado D.** & Rueda-Almonacid J.V. (2017) Phylogenetic Systematics of Dart-Poison Frogs and Their Relatives Revisited (Anura: Dendrobatoidea). *South American Journal of Herpetology* **12(s1)**: S1–S90. DOI: 10.2994/SAJH-D-17-00017.1.

2016

Jacob Machado D., Lyra M.L. & Grant, T. (2016) Mitogenome assembly from genomic multiplex libraries: Comparison of strategies and novel mitogenomes for five species of frogs. *Molecular Ecology Resources*, **16(3)**: 686–693. DOI: 10.1111/1755-0998.12492.

2015

Jacob Machado D. (2015) YBYRÁ facilitates comparison of large phylogenetic trees. *BMC Bioinformatics*, **16(1)**: 204. DOI: 10.1186/s12859-015-0642-9.

2012

Jacob Machado D. & Marques F.P.L. (2012) The forgotten origin of *Acanthobothrium* Blanchard, 1848 (Tetraphyllidea: Onchobothriidae). *Zootaxa* **3505**: 86–88. DOI: 10.11646/zootaxa.3505.1.7.

Under Review

de Paula Oliveira, H., **Jacob Machado, D.**, Prieto-Oliveira, P., & Ocaña, K. Evidence for recombination in dengue virus genomes. *Molecular Phylogenetics and Evolution*, manuscript MPE-D-26-00296. First submitted on June 17, 2026. The editor asked for minor revisions on October 8, 2026, and the second revision was submitted on October 21, 2026. Preprint DOI: 10.64898/2026.06.14.732057.

Marne, O., Bartman, B., Duchenko, A., & **Jacob Machado, D.** Alignment, outgroup, and branch statistic choices shape SARS-CoV-2 phylogenomics. *Frontiers in Bioinformatics*, section Evolutionary Bioinformatics, manuscript 1996052. Submitted on September 11, 2026. Two reviewer reports received, and the interactive review opened on October 25, 2026.

Preprints

de Paula Oliveira, H., **Jacob Machado, D.**, Prieto-Oliveira, P., & Ocaña, K. (2026). Evidence for recombination in dengue virus genomes. Preprint. DOI: 10.64898/2026.06.14.732057.

Terra, R., Carvalho, D., **Jacob Machado, D.**, Osthoff, C., & Ocaña, K. (2026). HP2NET: empowering efficient phylogenetic network analysis through high-performance computing. Preprint. DOI: 10.64898/2026.03.05.709005.

Duchenko, A., & **Jacob Machado, D.** (2026). AlphaFold and the transformation of structural biology: evolution, applications, limitations, and future directions. *Authorea Preprints*. DOI: 10.22541/au.177282022.20654724/v1.

Trevisan, B., **Jacob Machado, D.**, Lahr, D. J., & Marques, F. P. de L. (2024). Comparative characterization of mitogenomes from five orders of elasmobranch parasites indicate that total evidence analysis is superior to single genes for cestodes (Cestoda: Tapeworms). *Authorea Preprints*. DOI: 10.22541/au.170668376.62980776/v1. Posted after the same study appeared in *Frontiers in Genetics* (2021), listed above.

Yasa, S., Guirales-Medrano, S., **Jacob Machado, D.**, Ford, C.T., and Janies, D. (2023). Predicting antibody and ACE2 affinity for SARS-CoV-2 BA.2.86 with *in silico* protein modeling and docking. *bioRxiv*. DOI: 10.1101/2023.11.22.568364. Published in *Frontiers in Virology* (2024), listed above.

Ford C.T., Scott R., **Jacob Machado D.**, and Janies D.A. (2021) Sequencing data of North American SARS-CoV-2 isolates shows widespread complex variants. *medRxiv*. DOI: 10.1101/2021.01.27.21250648.

Karthigeyan K.P., Flanigan C., **Jacob Machado D.**, Kiziltas A.A., Janies D.A., Chen J., Cooke D., Lee M.V., Saif L.J., Henegar S., Jahnes J., Mielewski D.F., and Kwiek J.J. (2021) Heat efficiently inactivates coronaviruses inside vehicles. *bioRxiv*. DOI: 10.1101/2021.09.08.459486.

Patent applications

The provisional patent application titled "Compositions for Modulating Mutable Collagenous Tissue and Methods of Use Thereof" was filed with the United States Patent and Trademark Office (USPTO) on January 8, 2025, under Application No. 63/742,995. This Utility-Provisional Application was submitted under 35 USC 111(b) and is associated with Patent Center No. 68651418 and Attorney Docket No. 1276/12 PROV. The application was filed by Christopher Perkins, with the first-named inventor being Dr. Denis Jacob Machado, alongside co-inventors Ms. Reyhaneh Nouri, Dr. Vladimir Mashanov, and Dr. Daniel Janies. This innovation focuses on novel compositions and methods for modulating mutable collagenous tissue, with potential applications in biomaterials and regenerative medicine. On December 12, 2025, UNC Charlotte's Office of Research Commercialization and Development filed a non-provisional application from the provisional, which keeps the priority date of the January 8, 2025 filing.

Grants

Funded

Dec. 2024–Dec. 2026 **CNPq/MCTI: "ARISE in HPC: Artificial Intelligence for Recombination Identification and Surveillance in Epidemiology in a CAD Environment" (collaborator with shared supervision and methods development; PI and grantee: Kary Ocaña, LNCC)**, *Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)*, Brazil, process no. 444558/2024-1, call CNPq/MCTI/FNDCT No. 22/2024 – Programa Conhecimento Brasil
R$ 219.591,00

Jul. 2026–Jun. 2027 **Faculty Research Grant (FRG-1): "Single-Cell Sequencing of a Shape-Shifting Biological Material to Enable Smart Biomaterial Design" (PI)**, *UNC Charlotte, College of Computing and Informatics*, U.S.A., fund no. 101525
US$ 10,000.00

May–Aug., 2025 **CCI Dean's Summer Education Project: "Building AI-Readiness Across Bioinformatics Core Courses" (with Adam Whaley and Xiuxia Du)**, *University of North Carolina at Charlotte*, U.S.A., confirmed Apr. 18, 2025; paid as three stipends of US$ 3,333
US$ 10,000.00

Summer 2024 **Office of Undergraduate Research (OUR) Scholar Award, to support an undergraduate researcher in viral recombination studies**, *University of North Carolina at Charlotte*, U.S.A., internal award
US$ 5,000.00

2024–2025 **Student Success Grant: "Undo 'Stuck in the Shallow End': Charlotte's Interdisciplinary and Cross Campus CS/Computing Education Initiative for Women, Latino, and African American Students" (PI: Xiaoxia Newton; I was one of 11 team members)**, *University of North Carolina at Charlotte*, U.S.A., internal grant
US$ 15,000.00

Feb. 2024 **Free Expression and Constructive Dialogue Mini-Grant, for a campus Iftar community dinner (with Laurel Slane)**, *University of North Carolina at Charlotte*, U.S.A., internal grant
US$ 1,000.00

2024 **Academic Affairs one-time funding: "Boost UNC Charlotte's AI research with cost-effective, cutting-edge computing: AI-optimized machines with GPUs in the Research Computing environment" (with Daniel Janies and Benjamin Radford)**, *University of North Carolina at Charlotte*, U.S.A., internal request, submitted Jan. 8, 2024, and approved Feb. 2024, for 64 compute cores and 4 GPUs
US$ 80,000.00

2023–2024 **Research Internship Abroad (BEPE): "Mitogenome Organization and Diversity of Proteocephalid Tapeworms" (fellow: Philippe Vieira Alves)**, *São Paulo Research Foundation (FAPESP)*, Brazil, award no. 2023/00714-5
US$ 70,012.00

May 2023–Apr. 2025 **Mentored Pilot Award: "A Large Scale Virtual Screen of Compound Databases against Viral Drug Targets" (PI of the subaward; mentor: Daniel Janies)**, *National Institutes of Health (NIH/NIAID), subaward from the READDI-AViDD Center (U19) at UNC Chapel Hill*, U.S.A., award no. 1U19AI171292, subaward no. 5128868
US$ 462,000.00

Jan. 2023–May 2024 **Faculty Research Grant (FRG): "New horizons in regenerative medicine through the investigation of genes involved in the modulation of echinoderm mutable connective tissue (MCT)" (PI)**, *University of North Carolina at Charlotte (UNC Charlotte)*, U.S.A., fund no. 111139
US$ 8,000.00

Jul. 2023–Jun. 2025 **Ignite for Centers: CIPHER Research Center (Senior Personnel; PI: Daniel Janies)**, *University of North Carolina at Charlotte, Division of Research*, U.S.A., internal grant
Amount not specified

Mar. 2023–Feb. 2027 **REU Site: "Smart and Secure Future Computing" (Senior Personnel and Mentor)**, *National Science Foundation (NSF)*, award no. 2244424 (PI: Heather Lipford), US$ 368,685.00

2019–2022 **Young Researchers (2) Program (PI: Taran Grant; I collaborated as a bioinformatician and molecular biologist)**, *São Paulo Research Foundation (FAPESP)*, file no. 2018/15425-0, US$ 1,221,990.00

May 2018–Apr. 2022 **R15 AREA project to develop genomic and experimental resources to study regeneration (PI: Daniel Janies; I was a postdoctoral researcher on the award from 2019)**, *National Institutes of Health (NIH)*, file no. 1R15GM128066-01, US$ 461,250.00

2018–2020 **Regular Research Project from Continuous Funding Stream (PI: Fernando P. L. Marques; I collaborated as a bioinformatician and molecular biologist)**, *São Paulo Research Foundation (FAPESP)*, file no. 2018/03534-0, US$ 41,296.00

2016–2017 **Research Internship Abroad**, *São Paulo Research Foundation (FAPESP)*, file no. 2015/18654-2, US$ 26,080.00

2013–2018 **Funding of Doctorate Research Project: "Comparative Genomics and the Evolution of Amphibian Chemical Defense"**, *São Paulo Research Foundation (FAPESP)*, file no. 2013/05958-8, US$ 84,295.00

2010–2012 **Funding of Master's Research Project: "Phylogenetic diversity of the lineages of *Acanthobothrium* van Beneden, 1850 (Eucestoda: Tetraphyllidea: Onchobothriidae) parasites of Potamotrygonidae (Chondrichthyes: Myliobatoidei)"**, *São Paulo Research Foundation (FAPESP)*, file no. 2009/13561-5, US$ 19,101.00

May–Dec., 2009 **Funding for Scientific Initiation: "Phylogenetic position of phyllobothriids (Eucestoda: Tetraphyllidea) parasites of freshwater stingrays of the family Potamotrygonidae (Chondrichthyes: Myliobatoidei) based on direct optimization of nucleotide sequences"**, *São Paulo Research Foundation (FAPESP)*, file no. 2009/00886-3, US$ 1,477.00

Pending

Oct. 2026 **NIH R03: "Phyloinformatics of circadian onset: dating clock competence across vertebrates and predicting the human window" (contact PI, multiple PIs with Daniel Stanton)**, *National Institutes of Health (NIH), NICHD, PA-25-302*, U.S.A., submitted Oct. 16, 2026, for Jul. 2027 to Jun. 2029
US$ 154,000 (requested)

Jul. 2026 **NSF CAREER: "CAREER: Phyloinformatics for Predictive Evolutionary Inference" (PI)**, *National Science Foundation (NSF), Faculty Early Career Development Program*, U.S.A., proposal no. 2635288, submitted Jul. 22, 2026, for Jan. 2027 to Dec. 2031
US$ 942,760 (requested)

In prep.

2026 **AIDatasets:Flagship: "PhyloBank, an AI-Ready Evidence Layer for Phylogenetics" (with Adam Reitzel and Daniel Janies)**, *National Science Foundation (NSF 26-512)*, U.S.A., in preparation
Amount not specified

2026 **Collaborative proposal on the integrative taxonomy and phylogenetics of Passalidae (with Cristian Beza-Beza and Larry Jiménez-Ferbans, University of Minnesota)**, *National Science Foundation (NSF)*, U.S.A., in preparation
Amount not specified

2026 **Proposal on echinoderm mutable collagenous tissue (with Vladimir Mashanov)**, *National Science Foundation (NSF), Division of Integrative Organismal Systems (IOS)*, U.S.A., in preparation
Amount not specified

Submitted, Not Funded

2026 **Faculty Research Grant (FRG-2): "Channels, Enablers, and Barriers for University-Industry Interaction (UII) in Bioinformatics" (with Karla Liboreiro and Daniel Janies)**, *UNC Charlotte, College of Computing and Informatics*, U.S.A., not funded
US$ 5,000 (requested)

2025–2026 **NCInnovation pre-application: "Biology-Inspired AI-Driven Design of Novel Proteins for Biomaterials" (PI: Daniel Janies, with Vladimir Mashanov and Colby Ford)**, *NCInnovation*, North Carolina, U.S.A., pre-application accepted in 2025, not selected for the full proposal in 2026
Amount not specified

Oct. 2025 **Gambrell Faculty Fellowship: "Undo 'Stuck in the Shallow End': Charlotte's Interdisciplinary and Cross-Campus CS/Computing Education Initiative" (Co-PI; PI: Debra Smith)**, *University of North Carolina at Charlotte*, U.S.A., submitted Oct. 2025, not selected Nov. 2025
US$ 21,000 (requested)

2025 **AERA 2026 Annual Meeting proposal: "Undo 'Stuck in the Shallow End': A Mixed-Methods and Cross-Campus Interdisciplinary Approach to Computing Education" (lead: Xiaoxia Newton)**, *American Educational Research Association (AERA)*, U.S.A., submitted, not accepted

Sep. 2024 **Santander Research Internship: "Genomics of Endangered Insular Snakes from the *Bothrops jararaca* Group" (host for Igor Salles de Oliveira)**, *Santander Mobility, Discentes, University of São Paulo*, Brazil, Edital PRPG No. 11/2024
Amount not specified

2024 **Fulbright Scholar proposal: "Machine Learning to Study Dengue Virus Recombination in Pakistan and Worldwide" (with Muhammad Iqbal Qureshi)**, *Fulbright Program, U.S. Educational Foundation in Pakistan (USEFP)*, Pakistan/U.S.A., proposal no. FS/2025/45360
Amount not specified

May 2024 **NSF Research Traineeship (NRT) white paper for the UNC Charlotte limited-submission competition: "CCI EVOLVE: Training and Career Development in Computational Evolutionary Genomics" (Senior Personnel; leads: Rebekah Rogers and Way Sung)**, *National Science Foundation (NSF), NRT Program*, U.S.A., not selected by UNC Charlotte for submission to NSF, Jun. 2024
about US$ 3,000,000 (planned)

2024 **Division of Research Postdoctoral Hiring Program**, *University of North Carolina at Charlotte*, U.S.A., not funded

Dec. 2023 **NSF PIPP Phase II center proposal on preventing the pre-emergence of zoonotic pathogens (Co-Investigator and Aim 1 Lead, PI: Daniel Janies)**, *National Science Foundation (NSF), Predictive Intelligence for Pandemic Prevention (PIPP)*, U.S.A., submitted Dec. 8, 2023, declined May 2024
about US$ 18,000,000 (requested)

Oct. 2023 **"High-Performance Computing for Inferences of Viral Genome Recombination: Mechanisms, Methods and Evolutionary Consequences" (with Kary Ocaña)**, *Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)*, Brazil, positively reviewed but not funded, then revised and funded in 2024 as ARISE in HPC
about US$ 86,000

Feb. 2023 **NIH NIGMS R15 AREA renewal: "Genomic and Regulatory Mechanisms of Tissue Regeneration in Echinoderms" (Senior Personnel; PI: Daniel Janies)**, *National Institutes of Health (NIH)*, U.S.A., submitted Feb. 2023, not funded

Event Organization

2028 **Annual Meeting of the Willi Hennig Society (host)**, *UNC Charlotte*, Charlotte, NC, U.S.A.
• Description: I am bringing the Annual Meeting of the Willi Hennig Society to UNC Charlotte in 2028, with the support of the Department of Bioinformatics and Genomics.
• Society homepage: `https://cladistics.org/meetings/`

Jul. 23, 2025 **XLII Annual Meeting of the Willi Hennig Society**, *The Chinese University of Hong Kong, HKSAR*, Hong Kong, China
• Description: In July 2025, Dr. Jacob Machado organized and convened an international symposium titled "Phylogenetics & Public Health: Lessons from COVID-19 and Emerging Threats" at the XLII Annual Meeting of the Willi Hennig Society in Hong Kong. The symposium brought together international researchers to discuss applications of phylogenetics to public health, including SARS-CoV-2, mpox, H5N1, and papillomaviruses. Jacob Machado served as symposium organizer and moderator, opening and closing the session and coordinating speakers, scope, and scientific focus. The symposium emphasized methodological rigor, best practices in phylogenetic inference, and the role of evolutionary systematics in responding to emerging infectious disease threats.
• Event homepage: `https://cladistics.org/xlii-annual-meeting-2025`

Nov. 22, 2024 **Fall 2024 Bioinformatics Poster Showcase**, *UNC Charlotte, Bioinformatics Building*, Charlotte, NC, U.S.A.
• Description: The Bioinformatics Graduate Student Poster Showcase is an annual event where students from UNC Charlotte's Department of Bioinformatics present their research. As an organizer, I facilitated the event's planning, including coordinating student participation and logistics. Posters covered diverse bioinformatics topics, such as phylogenetics, AI applications in genomics, and molecular evolution.
• Event homepage: `https://phyloinformatics.com`

Apr. 23, 2024 **Charlotte HPC Group Workshop: "Workflow Automation with Snakemake and Nextflow"**, *UNC Charlotte, CIPHER Center (BINF 408)*, Charlotte, NC, U.S.A.
• Description: This workshop, hosted by the Charlotte HPC Group, introduced researchers to workflow automation in high-performance computing (HPC). It focused on Snakemake and Nextflow, two widely used tools for managing computational pipelines. Participants learned how to streamline data processing and improve reproducibility in bioinformatics and computational research.
• Event homepage: `https://cipher.charlotte.edu`

Jul. 11, 2023 **Symposium "Phylogenetics as Applied to Infectious Diseases" at the XL Annual Meeting of the Willi Hennig Society**, *Cornell University*, Ithaca, NY, U.S.A.
• Description: Daniel Janies and I organized this symposium. Speakers included Daniel Janies, Lucas Czech and Alexandros Stamatakis, Alex Dornburg and colleagues, Kary Ocaña and colleagues, and me.
• Society homepage: `https://cladistics.org/`

Jul. 7–28, 2022 **III Virtual Meeting of Systematics, Biogeography, and Evolution (SBE)**, *Online*, Worldwide
• Description: The SBE meeting 2022 will happen on every Thursday of July, 2022, from day 7 to day 28, on UTC-4. Each day will be composed of two symposia hosted over Zoom and one virtual poster session in the SBE meeting 2022's Discord server. The SBE meetings are scientific events organized by mostly young Latinx researchers under a zero budge, zero profit, and zero cost strategy.
• Event's homepage: `https://www.sbemeeting.weebly.com`

Jun. 19–23, 2021 **II Virtual Meeting of Systematics, Biogeography, and Evolution (SBE): The Research of Biodiversity and the Diversity of Researchers**, *Online*, Worldwide
• Description: The SBE meeting 2021's primary purpose was to reach people who usually have economic restrictions to attend most of the scientific meetings on different areas within evolutionary biology. In its second edition, the event had 10 symposia, 1 workshop, and over 2,000 attendees from hundreds of institutions worldwide.
• Event's homepage: `https://www.sbemeeting.com`

Jul. 28–30, 2020 **I Virtual Meeting of Systematics, Biogeography, and Evolution (SBE): A Joint Effort in the Coronavirtual Era**, *Online*, Worldwide

• Description: The SBE meeting is a space for socialization and scientific dialogue that promotes the advancement of the discipline and encourages inclusive participation based on gender equality in all conferences and symposia. In its first edition, it had over 1,200 scientists enrolled from 38 countries
• Event's homepage: `https://sbemeeting.weebly.com`

Oral Presentations

September 14, 2026 **Charlotte Regional Biomedical Symposium**, *Charlotte, NC*, USA
• Title: The SARS-CoV-2 phylogeny is sensitive to alignment parameters and outgroup selection
• Authors: Omkar Marne and Jacob Machado, D.
• Format: Oral presentation (presented by Omkar Marne)

July 10, 2026 **EVOLVE program, UNC Charlotte**, *Charlotte, NC*, USA
• Topic: Péva, the lab's phylogenetics platform
• Authors: Jacob Machado, D.
• Format: Oral presentation

June 8, 2026 **GRADS-4C Research Symposium "AI and Digital Health: Potential, Promise, and Peril"**, *Greensboro, NC*, USA
• Title: FAIR Docking: Best practices for reproducible protein-protein interactions on cloud platforms
• Authors: Anastasiia Duchenko and Jacob Machado, D.
• Format: Rapid talk (presented by Anastasiia Duchenko)
• Host: Genomics Research and Data Science Center for Computation and Cloud Computing (GRADS-4C), North Carolina A&T State University

May 14, 2026 **ThinkAI Research and Innovation Symposium**, *Charlotte, NC*, USA
• Title: Promoting reproducible research with AVOIDRUNE: An automated pipeline for protein-protein interaction analysis
• Authors: Anastasiia Duchenko and Jacob Machado, D.
• Format: Oral presentation (presented by Anastasiia Duchenko)

April 17, 2026 **Larry Mays Seminars in Bioinformatics, UNC Charlotte**, *Charlotte, NC*, USA
• Title: The Phyloinformatics Club Special Meeting on Museomics
• Authors: Jacob Machado, D.
• Format: Seminar speaker

April 7, 2026 **UNC Charlotte Graduate Research Symposium (GRS)**, *Charlotte, NC*, USA
• Title: The SARS-CoV-2 phylogeny is sensitive to alignment parameters and outgroup selection
• Authors: Omkar Marne and Jacob Machado, D.
• Format: Oral presentation (presented by Omkar Marne)

December 17, 2025 **Immunology, Microbiology, and Virology Focus Group, Department of Biological Sciences, UNC Charlotte**, *Charlotte, NC*, USA
• Title: Back to Basics: Phylogenetic Principles for Robust Viral Analysis
• Authors: Jacob Machado, D.
• Format: Seminar

November 20, 2025 **EVOLVE program, UNC Charlotte**, *Charlotte, NC*, USA
• Topic: Alignments and trees
• Authors: Jacob Machado, D.
• Format: Session

July 24, 2025 **XLII Annual Meeting of the Willi Hennig Society**, *The Chinese University of Hong Kong*, Hong Kong SAR, China
- Title: Beetles, bytes, and branches: Integrating artificial intelligence into the taxonomy and phylogenetics of Passalidae
- Authors: Larry Jiménez-Ferbans, Cristian Beza-Beza, Anastasiia Duchenko, Aram Mikaelyan, and Jacob Machado, D.
- Format: Oral presentation
- Conference website: `https://willi-hennig-society.org/meetings/2025`

July 23, 2025 **XLII Annual Meeting of the Willi Hennig Society**, *The Chinese University of Hong Kong*, Hong Kong SAR, China
- Title: Advancing phylogenetics for health threats: Lessons from the COVID-19 pandemic and beyond
- Authors: Omkar Marne and Jacob Machado, D.
- Format: Oral presentation (Symposium 1: Phylogenetics & Public Health)
- Conference website: `https://willi-hennig-society.org/meetings/2025`

June 7, 2025 **100th Annual Meeting of the American Society of Parasitologists**, *Winston-Salem, NC*, USA
- Title: AI-driven predictions of phylogenetic trees from zoogeographical data in dactylogyrids (Platyhelminthes: Monogenea)
- Authors: Jacob Machado, D., Aline A. Acosta, and Anastasiia Duchenko
- Format: Oral presentation
- Conference website: `https://aspmeetings.wixsite.com/2025`

May 29, 2025 **Society for the Preservation of Natural History Collections (SPNHC) Annual Meeting**, *Lawrence, KS*, USA
- Title: Simplifying specimen loans: Ten simple rules for researchers
- Authors: Giovanna Yumi Scorsim Omura and Jacob Machado, D.
- Format: Oral presentation
- Conference website: `https://spnhc2025.ku.edu/`

April 17, 2025 **UNC Charlotte Undergraduate Research Conference (URC)**, *Charlotte, NC*, USA
- Title: FLAVi 2: Increasing Efficiency and Accessibility of Flaviviridae Genome Annotation
- Authors: Cayden Jacobsen and Jacob Machado, D.
- Format: Oral presentation
- Video: `https://www.youtube.com/watch?v=k6bCMXNy_NQ`

March 27, 2025 **Postdoctoral Readiness to Enter the Professorship Program (PREPP)**, *UNC System (online)*, USA
- Title: Preparing for the Faculty Interview
- Authors: Jacob Machado, D., Skylar Hopkins, Ebony Jones, and Maria Rodgers
- Format: Invited panel presentation
- Program homepage: `https://grad.ncsu.edu/professional-development/opa/professional-development/`

January 5, 2025 **Annual Meeting of the Society for Integrative and Comparative Biology (SICB)**, *Atlanta, GA*, USA
- Title: Unveiling the modulators of mutable collagenous tissue in the brittle star *Ophiomastix wendtii*: An RNA-Seq analysis
- Authors: Reyhaneh Nouri, Vladimir Mashanov, and Jacob Machado, D.
- Format: Oral presentation
- Slides: `https://gitlab.com/phyloinformatics-presentations/20241121/-/blob/main/reyhaneh_nouri/20250105_Nouri.pdf`
- Abstract: *Integrative and Comparative Biology*, **65**, S380 (2025)

October 21, 2024 **10th Annual Biological Sciences Research Symposium**, *Charlotte, NC*, USA
- Title: How much should we worry about SARS-CoV-2 mutations according to AI?
- Authors: Anastasiia Duchenko, under the supervision of Jun-tao Guo and Jacob Machado, D.
- Format: Oral presentation (presented by Anastasiia Duchenko)

September 9, 2024 **2024 Biomedical Sciences Symposium**, *Charlotte, NC*, USA
- Title: Unveiling the modulators of mutable collagenous tissue in the brittle star *Ophiomastix wendtii*: An RNA-Seq analysis
- Authors: Reyhaneh Nouri, Vladimir Mashanov, and Jacob Machado, D.
- Format: Oral presentation (presented by Reyhaneh Nouri)

July 30, 2024 **XLI Annual Meeting of the Willi Hennig Society**, *Mexico City*, Mexico
- Title: Twenty years after: What larval characters are still telling us about the phylogeny of frogs (Amphibia: Anura)
- Authors: Pedro Henrique dos Santos Dias, Alexander Haas, and Jacob Machado, D.
- Format: Oral presentation (presented by Jacob Machado, D.)
- Zenodo: 10.5281/zenodo.13126486

June 14–17, 2024 **99th Annual Meeting of the American Society of Parasitologists (ASP)**, *Denver, CO*, USA
- Title: Mitogenome organization, diversity, and evolutionary relationships of proteocephalidean tapeworms (Cestoda, Onchoproteocephalidea) unveiled by a genome skimming approach
- Authors: Philippe Vieira Alves, Reinaldo J. da Silva, Daniel Janies, William Taylor, April Harris, Gari New, and Jacob Machado, D.
- Zenodo: `https://zenodo.org/records/11477509`

— **99th Annual Meeting of the American Society of Parasitologists (ASP)**, *Denver, CO*, USA
- Title: Machine learning models accurately predict clades of proteocephalidean tapeworms (Onchoproteocephalidea) based on host and biogeographical data
- Authors: Philippe Vieira Alves, Reinaldo J. da Silva, Alain de Chambrier, José L. Luque, Anastasiia Duchenko, Daniel Janies, and Jacob Machado, D.
- Zenodo: `https://zenodo.org/records/11307234`

June 11–14, 2024 **X-Meeting 2024, Brazilian Bioinformatics Conference**, *Salvador, BA*, Brazil
- Title: Evidence of recombination in dengue virus (DENV)
- Authors: Hugo de Paula Oliveira, Paula Prieto Oliveira, Jacob Machado, D., and Kary Ocaña
- Format: Oral presentation

March 22, 2024 **UNC Charlotte 23rd Annual Graduate Research Symposium**, *Charlotte, NC*, USA
- Title: How can we reversibly control collagen pliability? An investigation of mutable collagenous tissues in the brittle star *Ophiomastix wendtii*
- Authors: Reyhaneh Nouri, Vladimir Mashanov, and Jacob Machado, D.
- Format: Oral presentation (presented by Reyhaneh Nouri)

June 15, 2023 **Calima Foundation**, *Cali*, Colombia
- Title: What is phyloinformatics and how can it help us respond faster to health emergencies?
- Authors: Jacob Machado, D.
- Format: Invited talk, at the invitation of Jhon Ospina-Sarria

2023 **XL Annual Meeting of the Willi Hennig Society**, *Ithaca–NY*, USA
- Title: Combining phylogenetic and artificial intelligence to unravel viral recombination
- Authors: Jacob Machado, D.
- Organization's homepage: `https://cladistics.org/`

— **Calima Foundation**, *Cali–Valle del Cauca*, Colombia
- Title: What is phyloinformatics and how can it help us respond faster to health emergencies?
- Authors: Jacob Machado, D.
- Organization's homepage: `https://en.fundacioncalima.org/`

— **UNC Charlotte's 9th Annual Biological Sciences Research Symposium**, *Charlotte-NC*, USA
- Title: How can we reversibly control collagen pliability? An investigation of mutable collagenous tissue in the brittle star *Ophiomastix wendtii*
- Authors: Nouri R., Mashanov V., and Jacob Machado D.

— **UNC Charlotte's 9th Annual Biological Sciences Research Symposium**, *Charlotte-NC*, USA
- Title: A phylogenomic and machine learning framework for discriminating SARS-CoV-2 variants
- Authors: Marne O. and Jacob Machado D.

— **The 7th International Conference on Interdisciplinary Research on Computer Science, Psychology, and Education (ICICPE–2023)**, *Bali*, Indonesia
- Title: Replicating Ford et al.'s investigation on the immune consequences and SARS-CoV-2 antibody's binding affinity using an in silico approach
- Authors: Jones C., Zeru H., Jacob Machado, D.
- Event's homepage: `http://icicpe.org/`
- Abstarct: ISSN 2508-7436, pages 246–250.
- Zenodo: `https://zenodo.org/records/10138987`

2022 **CCI Academic Research Seminar**, *Charlotte–NC*, USA
- Title: What is phyloinformatics and how can it help us respond faster to health emergencies?
- Authors: Jacob Machado D.
- Date: September 16, 2022
- Organization: College of Computing and Informatics, UNC Charlotte

— **Willi Hennig Society, Breaking Isolation lecture series (invited speaker)**, *Online*, Worldwide
- Topic: Phylogenetics and the One Health approach
- Authors: Jacob Machado D.
- Date: April 28, 2022, in a session shared with Pablo A. Goloboff
- Organizers: Christiane Weirauch and Dolores Casagranda

— **VI Symposium of Zoological Systematics**, *Online*, Worldwide
- Title: The role of phylogeneticists in the execution of the One Health approach.
- Authors: Jacob Machado D., Antunes, E.P., Omura, G.S.Y., Yohe L.
- Organization: Graduate Program in Zoology, Federal University of Minas Gerais
- Event's homepage: `https://szsufmg.wixsite.com/vi-szs-ufmg`
- Zenodo: `https://doi.org/10.5281/zenodo.6528875`

— **XV Academic Meeting of Computational Modeling**, *Online*, Worldwide
- Title: Análises filogenômica de coronavírus: uma discussão metodológica.
- Authors: Jacob Machado D.
- Organization: Summer Program 2022, National Laboratory of Scientific Computing (LNCC)

2021 **Special Symposium of UNC Charlotte's Bioinformatics Research Center**, *Online*, Worldwide
- Title: The origins of human coronaviruses.
- Authors: Janies D.A., Jacob Machado D., Scott R. & Guirales S.
- Organization: II Virtual Meeting of Systematics, Biogeography and Evolution: A Joint Effort in the Coronavirtual Era
- Event's homepage: `https:/www.sbemeeting.com`

— **Seminar Series**, *Santa Marta, Colombia*, Universidad del Magdalena
- Title: Epidemiología genómica y lecciones aprendidas de la pandemia de COVID-19.
- Authors: Jacob Machado D.
- Organization: Universidade del Magdalena

— **CARLA 2021**, *Online*, Worldwide
- Title: The HPC bottlenecks in the genomic surveillance of SARS-CoV-2.
- Authors: Jacob Machado D.
- Organization: Latin American High Performance Computing Conference
- Website: `carla2021.org`

— **Projetos COVID**, *Online*, Worldwide
- Title: The origins of human coronaviruses.
- Authors: Jacob Machado D.
- Organization: Hospital Israelita Albert Einstein

2020 **Virology in the SARS-CoV-2 Era**, *Online*, Worldwide
- Title: Phylogenomics of *Orthocoronavirinae*: evolutionary relationships between coronaviruses and their hosts.
- Authors: Jacob Machado D. & Janies D.
- Organization: I Virtual Meeting of Systematics, Biogeography and Evolution: A Joint Effort in the Coronavirtual Era
- Event's homepage: `https://sbemeeting.weebly.com/`

— **Phylopizza**, *Washington, D.C.*, Smithsonian Institution
- Title: What do poison dart frogs have to do with COVID-19?
- Authors: Jacob Machado D.
- Organization: Laboratory of Data Sciences, Smithsonian Institution

— **UNC Charlotte Continuing Education Webinar Series**, *Charlotte–NC*, USA
- Title: COVID-19: Understanding the Science Behind the Pandemic
- Authors: Janies D. & Jacob Machado D.
- Organization: UNC Charlotte Continuing Education and UNC Charlotte College of Computing and Informatics
- Recording available at `https://youtu.be/VmdNGyFRu3c`

2019 **XXXVIII Annual Meeting of the Willi Hennig Society**, *Berkeley–CA*, USA
- Title: Facing long-branch anxiety and outgroup prejudice in the phylogenomic analysis of *Flaviviridae*
- Authors: Jacob Machado D., Schneider A. de. B. & Janies D.
- Support: Department of Bioinformatics and Genomics, UNC Charlotte

— **XXXVIII Annual Meeting of the Willi Hennig Society**, *Berkeley–CA*, USA
- Title: Eclecticism in cladistics: convergence among opposing optimality criteria may be a hoax
- Authors: Jiménez-Ferbans L., Jacob Machado D. & de Freitas G.M.I.
- Support: Department of Bioinformatics and Genomics, UNC Charlotte

— **The 94th Annual Meeting of the American Society of Parasitologists**, *Rochester–MN*, USA
- Title: New dedicated pipeline to annotate the cestode mitogenome
- Authors: Trevisan B., Jacob Machado D., Marques F.P. de L.
- Support: Department of Bioinformatics and Genomics, UNC Charlotte

2017 **5th International Quest for Orthologs Meeting**, *Los Angeles–CA*, USA
- Title: Flavivirus phylogeny revisited: in search of the orthologs
- Autores: Schneider A. de. B., Jacob Machado D., Lambodhar D. & Janies D.
- Support: FAPESP & Department of Bioinformatics and Genomics, UNC Charlotte

2016 **35th Annual Meeting of the Willi Hennig Society and XII Reunión Argentina de Cladística y Biogeografia**, *Buenos Aires*, Argentina
- Title: Evidence of absence treated as absence of evidence: the effects of gaps in standart maximum likelihood analysis
- Authors: Jacob Machado D., Castroviejo-Fisher S. & Grant T.
- Support: FAPESP and UNC Charlotte

— **35th Annual Meeting of the Willi Hennig Society and XII Reunión Argentina de Cladística y Biogeografia**, *Buenos Aires*, Argentina
- Title: direct measures of support for maximum likelihood
- Authors: Jacob Machado D., Marques F.P. de L. & Grant T.
- Support: FAPESP and UNC Charlotte

2014 **X Congreso Latinoamericano de Zoologia**, *Cartajena das Indias*, Colombia
- Title: Reconstructing mitochondrial genomes for five species of amphibians directly from genomic next-generation sequencing reads
- Authors: Jacob Machado D., Lyra M.L. & Grant T.
- Support: FAPESP

— **X Congreso Latinoamericano de Zoologia**, *Cartajena das Indias*, Colombia
- Title: Phylogenetic analysis of transformation series composed of ordered sequences
- Authors: Dias P.H.S. & Jacob Machado D.
- Support: FAPESP

— **"Curso de Verão em Bioinformática"**, *University of São Paulo*, São Paulo–SP, Brazil
- Event organizer and panelist

2013 **XXXII Willi Hennig Meeting**, *University of Rostock*, Rostock, Germany
- Title: On the use of iterative pass as a refinement strategy
- Authors: Jacob Machado D. & Marques F.P. de L.
- Support: FAPESP

2011 **XXX Willi Hennig Meeting**, *Universidade Estadual Paulista "Júlio de Mesquita Filho"*, São José do Rio Preto–SP, Brazil
- Title: Phylogenetic position of phyllobothriids (Eucestoda: Tetraphyllidea: Phyllobothriidea) parasites of Neotropical freshwater stingrays (Chondrichthyies: Myliobatoidei: Potamotrygonidae) based on direct optimization of nucleotide sequences
- Authors: Jacob Machado D. & Marques F.P. de L.
- Support: FAPESP

— **7th International Workshop on Cestode Systematics**, *University of Kansas*, Lawrence–KS, USA
• Title: Defining species boundaries using integrated taxonomy for freshwater tetraphyllideans: How to recognize and deal with great morphological variation?
• Authors: Marques F.P. de L., Reyda F.B., Prado P.I., Bueno V.M., Jacob Machado D. & Luchetti N.M.
• Support: FAPESP

Poster Presentations

November 19, 2026 **Evaluation 2026, American Evaluation Association (AEA)**, *Charlotte, NC*, USA
• Title: Undo "Stuck in the Shallow End": A Mixed-Methods and Cross-Campus Interdisciplinary Approach to Computing Education
• Authors: Xiaoxia Newton, Jacob Machado, D., Nick Findlater, Jing Yang, Debra Smith, and Courtney Green
• Format: Poster presentation (accepted July 15, 2026, AEA proposal 2419618)

October 1, 2026 **71st Brazilian Congress of Genetics (Genética 2026)**, *Florianópolis, SC*, Brazil
• Title: Morphological and genomic characterization of a novel *Hepatozoon* lineage in the critically endangered insular pitviper *Bothrops sazimai* (Squamata: Serpentes: Viperidae)
• Authors: Giselle Pessanha Pessoa, Igor Salles de Oliveira, Giovanna Yumi Scorsim Omura, Jacob Machado, D., Milton Yutaka Nishiyama Junior, and Maria José de Jesus Silva
• Format: Poster presentation (presented by Giselle Pessanha Pessoa)
• Poster and source: `https://github.com/machadodj/CBG71_Hepatozoon`

September 30, 2026 **71st Brazilian Congress of Genetics (Genética 2026)**, *Florianópolis, SC*, Brazil
• Title: Evolution of the repeatome and satellitome in three endangered insular species of the genus *Bothrops* (Squamata: Serpentes: Viperidae)
• Authors: Igor Salles de Oliveira, Giselle Pessanha Pessoa, Giovanna Yumi Scorsim Omura, Jacob Machado, D., Milton Yutaka Nishiyama Junior, and Maria José de Jesus Silva
• Format: Poster presentation (presented by Igor Salles de Oliveira)
• Poster and source: `https://github.com/machadodj/CBG71_Snakes`

July 20, 2026 **Summer Research Symposium 2026, Office of Undergraduate Research**, *Charlotte, NC*, USA
• Title: Péva has the first automated tool for measuring the association between a phylogeny and its metadata
• Authors: Emily Biaz and Jacob Machado, D.
• Support: REU Site "Smart and Secure Future Computing" (NSF #2244424)
• Materials: `https://gitlab.com/phyloinformatics/peva-public/-/tree/main/research/reu26/biaz_hierarchical_clade_prediction`

— **Summer Research Symposium 2026, Office of Undergraduate Research**, *Charlotte, NC*, USA
• Title: Benchmarking Péva against RDP5 for viral recombination detection using simulated alignments
• Authors: Sumedha Budaraju and Jacob Machado, D.
• Support: REU Site "Smart and Secure Future Computing" (NSF #2244424)
• Materials: `https://gitlab.com/phyloinformatics/peva-public/-/tree/main/research/reu26/budaraju_recombination_benchmark`

April 23, 2025 **Mutations in Space and Time Conference**, *Broad Discovery Center, Cambridge, MA*, USA
• Title: Automated Protocol for Testing Crucial Mutations in Coronaviruses Using AI: Insights into Immune Escape and Resistance
• Authors: Anastasiia Duchenko, Daniel Janies, and Jacob Machado, D.
• Format: Poster presentation
• Project page: `https://gitlab.com/staisyelash/2025_mutationsintimeandspace_duchenko`

November 22, 2024 **Fall 2024 Bioinformatics Poster Showcase**, *Charlotte, NC*, USA
- Anastasiia Duchenko: SARS-CoV-2 mutations in Spike RBD and its impact on the docking to antibodies and ACE2
- Prudvi Ragi: Investigating sodium voltage-gated channel alpha subunit 4 (SCN4A) mutations in poison frogs
- Shivam Patel: Docking analyses to identify mutations in the sodium channels of poison dart frogs that may confer resistance against lipophilic alkaloids
- Tulasi Arla: New software for recombination simulation and phylogenomic character categorization
- Vedika Judyani: Viral recombination detection using machine learning
- Materials: `https://gitlab.com/phyloinformatics-presentations/20241121`

July 26, 2024 **REU Site "Smart and Secure Future Computing", College of Computing and Informatics**, *Charlotte, NC*, USA
- Title: Training a natural language processing tool to parse and standardize species descriptions and create searchable databases to assist taxonomists on Passalidae (bess beetles) identification: preliminary data using ChatGPT
- Authors: Amariah Robinson and Jacob Machado, D.
- Support: REU Site "Smart and Secure Future Computing" (NSF #2244424)
- Materials: `https://gitlab.com/phyloinformatics/passalids/-/tree/main/chatbot`

— **REU Site "Smart and Secure Future Computing", College of Computing and Informatics**, *Charlotte, NC*, USA
- Title: Leveraging convolutional neural networks (CNNs) for accurate identification of Passalidae species to aid ecological research and global warming
- Authors: Manish Kota and Jacob Machado, D.
- Support: REU Site "Smart and Secure Future Computing" (NSF #2244424)
- Materials: `https://gitlab.com/phyloinformatics/passalids/-/tree/main/image_classification`

June 11–14, 2024 **20th X-Meeting, Brazilian Congress of Bioinformatics**, *Salvador, Bahia*, Brazil
- Title: Recombinant analyses in dengue virus type 1 with RDP5
- Authors: Hugo de Paula Oliveira, Paula Prieto Oliveira, Jacob Machado, D., and Kary Ocaña
- Event's homepage: `https://www.even3.com.br/xmeeting-2024/`
- Poster available at: `https://zenodo.org/records/11529272`

July 28, 2023 **REU Site "Smart and Secure Future Computing", College of Computing and Informatics**, *Charlotte, NC*, USA
- Title: Replicating Ford *et al.*'s investigation on the immune consequences and SARS-CoV-2 antibody's binding affinity using an *in silico* approach
- Authors: Cameron Jones, Hannah Zeru, and Jacob Machado, D.
- Materials: `https://drive.google.com/drive/folders/1iPcip7kIEQxjv9oHRKLypa_zOFGditk-?usp=sharing`

April 28, 2023 **Spring Graduate Poster Session, Department of Bioinformatics and Genomics**, *Charlotte, NC*, USA
- Title: Are there alkaloid-resistance mutations in *Phyllobates terribilis*' sodium channels?
- Authors: Reyhaneh Nouri and Jacob Machado, D.

— **Spring Graduate Poster Session, Department of Bioinformatics and Genomics**, *Charlotte, NC*, USA
- Title: A phylogenomics and machine learning framework for discriminating SARS-CoV-2 variants
- Authors: Omkar Marne and Jacob Machado, D.

2020 **ASBMB Annual Meeting**, USA
- FLAVi-Web: A Web Annotator for Viral Genomes of Flaviviridae with a Revised Phylogeny of the Family
- Authors: Guirales S., Jacob Machado D., Schneider A.de B. & Janies D.
- Support: Department of Bioinformatics and Genomics, UNC Charlotte
- Publication: The Federation of American Societies for Experimental Biology (FASEB) Journal
- DOI: 10.1096/fasebj.2020.34.s1.09730

2019 **XXXVIII Annual Meeting of the Willi Hennig Society**, *Berkeley–CA*, USA
- Enhanced mitogenome annotation and phylogenetic analysis of Cestode
- Authors: de Freitas G.M.I. & Jacob Machado D.
- Support: Department of Bioinformatics and Genomics, UNC Charlotte

2018 **Virus Genomics and Evolution**, *Cambridge*, UK
- Stopping to compare apples and oranges: a homology-based phylogeny of Flaviviridae
- Authors: Schneider A. de. B. & Jacob Machado D.

2013 **XXXII Willi Hennig Meeting**, *University of Rostock*, Rostock, Germany
- YBYRÁ - a fast and resourceful tool for examining clade prevalences in large sets of trees
- Authors: Jacob Machado D. & Marques F.P. de L.
- Support: FAPESP

Teaching Experience

Graduate School

2024–current **Instructor (Faculty)**, *University of North Carolina at Charlotte (UNC Charlotte)*, Bioinformatics Programming II (BINF-6112/8112), 3 credits (Spring & Fall)
- Shell scripting and high-performance computing training
- Python programming for handling biological data

Oct. 2026 **Workshop Instructor**, *University of North Carolina at Charlotte (UNC Charlotte), EVOLVE*, Two hands-on workshops on Péva, 2 sessions, 1.5 hours each
- Workshop 1 (October 1): phylogenetic analysis with Péva
- Workshop 2 (October 8): analyzing results with Péva
- Free and open materials: slide deck, written guide, data, and one script per step

2022, Fall **Instructor (Faculty)**, *University of North Carolina at Charlotte (UNC Charlotte)*, "Machine Learning for Bioinformatics" (BINF-6210/8210), 3 credits
- Theory and practice of machine learning, artificial neural networks, and deep learning, with focus on bioinformatics applications.

2021 **Workshop Instructor**, *Penn State University*, "Don't panic: a survival guide to RNA-Seq", 16 hours, single event
- Theory and practice on RNA-Seq and gene expression analysis for biomedical applications .

2019 **Invited Lecturer**, *University of São Paulo*, "Introduction to programming for biologists" (BIZ5788), 60 hours
- Taught lecture and lab to 20 graduate students and postdocs
- Evaluated highly by students for introducing original course material that teaches Bash, Python, and R using examples that integrate different fields, from phylogenetics to systems biology

2018 **Invited Lecturer**, *University of São Paulo*, "Introduction to bioinformatics: a theory-practice course with examples from the phylogenomics of *Flavivirus*" (BIZ5785), 60 hours
- Taught lecture and lab to 40 graduate students and postdocs
- Developed bioinformatics lab exercises based on open-source tools
- Evaluated highly by students for promoting critical thinking
- Introduced hack-a-thon events that helped solving local problems
- Promotes learning interdisciplinary concepts and practicing bioinformatics following the step-by-step protocols of a real and original research paper

2017 & 2025 **Invited Lecturer**, *University of Magdalena*, Introduction to bioinformatics (2017) and "Bioinformatics for Non-Bioinformaticists" (June 9–13, 2025), 40 hours each
- Taught lecture and lab to over 20 students, including undergraduate and graduate students, postdocs, staff, and faculty
- Raised resources to expand the local library and computational equipment

College Level

2025–current **Instructor (Faculty)**, *University of North Carolina at Charlotte (UNC Charlotte)*, Intro to Computer Science II (ITSC1213-001), 3 credits (Summer)
- Best practices in computer programming
- Git and GitHub
- Java programming

Aug – Dec 2013 **Teacher Assistant**, *University of São Paulo*, Department of Zoology
- Teaching: Vertebrate Zoology
- Supervissor: Professor Taran Grant

Aug – Dec 2011 **Teacher Assistant**, *University of São Paulo*, Department of Zoology
- Teaching: Diversity and Biogeography of the Neotropical Fauna
- Supervisor: Professor Ricardo Pinto da Rocha

Mar – Jul 2011 **Teacher Assistant**, *Universidade de São Paulo*, Department of Zoology
- Teaching: Foundations of Systematics and Biogeography
- Supervisor: Professor Fernando Portella de Luna Marques

Aug – Dec 2010 **Teacher Assistant**, *São Paulo State University*, Marine Biology
- Teaching: Invertebrate Zoology II
- Supervisor: Professor Tânia Márcia Costa

K-12

2018 – 2019 **Science Teacher**, *Kindy Escola Americana*, São Paulo–SP, Brazil
- Responsible for the science courses
- Developed and led hands-on activities
- Practice strategies to promote gender and social equity

2011 – 2013 **Informatics Instructor**, *Instituto Brasileiro de Formação e Capacitação* (IBFC), Taboão da Serra–SP, Brazil
- Taught lecture and lab on computer history, Microssoft application, Email, and Cloud Computing
- Non-governmental agency aiming to offer opportunities in low-income communities

2011 – 2012 **Coordinator of Exact Sciences**, *Oficina do Estudante*, São Paulo–SP, Brazil
- Taugh calculus, analytical geometry, basic physics, and basic chemistry
- Classes aimed to increase the chances of students from low-income families get access to a college education

Mentoring

The Phyloinformatics Lab currently has four Ph.D. candidates from the Bioinformatics and Computational Biology (BCB) Ph.D. Program: Anastasiia Duchenko, Omkar Marne, Reyhaneh Nouri, and Giovanna Yumi Scorsim Omura. The lab also hosts one undergraduate researcher, Srikritha Kosuri.

Please visit phyloinformatics.com/people for current information on my current team members and alumni.

2022–2026 **Alumni**, *Phyloinformatics Lab*, UNC Charlotte, Charlotte–NC, USA
- Paula Prieto Oliveira (*postdoc*)
- Philippe Vieira Alves (*postdoc*, placed first in the 2026 competition for a faculty position at São Paulo State University, UNESP)
- Isha Jain (*Master's*)
- Prudvi Raj Ragi (*Master's*)
- Shivam Patel (*Master's*)
- Tulasi Arla (*Master's*)
- Vedika Judyani (*Master's*)
- Drashti Mehta (*Master's*)
- Bou Haidara (*Master's*)
- Hugo de Paula Oliveira (*Master's*)
- Johannes Schultheis (*Master's*)
- Pavan Kalyan Reddy Konudula (*Master's*)
- Amariah Robinson (*undergrad*)
- Manish Kota (*undergrad*)
- Cayden Jacobsen (*undergrad*, Departmental Honors thesis on FLAVi 2, Spring 2025)
- Hannah Zeru (*undergrad*)
- Cameron Jones (*undergrad*)
- Melissa Sagastume (*undergrad*)
- Emily Biaz (*undergrad*, REU 2026)
- Sumedha Budaraju (*undergrad*, REU 2026)

2022–2026 **Trainee Awards**, *Phyloinformatics Lab*, UNC Charlotte, Charlotte–NC, USA
- Anastasiia Duchenko: Lawrence E. Mays Fellowship in Bioinformatics (2026–2027), Graduate Life Fellow (2026–2027), GRADS-4C Research Symposium Travel Award (2026), and GPSG Travel Funding
- Omkar Marne: Graduate School Summer Research Fellowship, and Honorable Mention for the Susumu Ohno Award (EVOLVE)
- Reyhaneh Nouri: William Ribarsky Memorial Scholarship (2026–2027), Graduate School Summer Research Fellowship, Atkins Library Open Access Publication Award, Thomas L. Reynolds Center for Graduate Life & Learning Fellowship, Lawrence E. Mays Fellowship in Bioinformatics, and International Student Scholarship
- Giovanna Yumi Scorsim Omura: Campus Superhero Award (Thomas L. Reynolds Center for Graduate Life & Learning), and Bioinformatics Assembly of Students Travel Award

2024–2026 **Graduate Committee Member**, *University of North Carolina at Charlotte*, Charlotte–NC, USA
- Yunfei (Eric) Liao, Ph.D. (advisor: Elizabeth Cooper), defended on April 2, 2025
- Dissertation: "Harnessing Computational Tools and Complex Biological Data for Environmental and Health Applications"
- Sayal Guirales-Medrano, Ph.D. (advisor: Daniel Janies), defense scheduled for October 29, 2026
- Melissa Hatley, Ph.D. in Biology (advisor: Bao-Hua Song), Graduate Faculty Representative from September 2024, defended on April 3, 2026

2025 **Thesis Supervisor**, *Universität Münster*, Münster, Germany
- Johannes Schultheis, M.Sc. in Biosciences, defended on March 20, 2025
- Thesis: "A Newt Hope: Combining RNA-Seq and Machine-Learning to Unravel the Evolution of Tetrodotoxin (TTX) defense in *Taricha granulosa*"
- Examiners: Joachim Kurtz (Universität Münster) and Taran Grant (Universidade de São Paulo)

Dr. Jacob Machado has extensive experience mentoring undergraduate and graduate students prior to the Phyloinformatics Lab.

2021–2022 **College of Computing and Informatics**, *University of North Carolina at Charlotte*, Charlotte–NC, USA
- Level: Doctorate (Projects 1 and 2) and Undergraduate (Projects 3 and 4)
- No. of students: 4
- Project 1: "Evolution of antimicrobial resistance in *Escherichia coli*"
- Project 2: "Improving genomic resources for highly regenerative echinoderms"
- Project 3: "Annotation of the draft genome of the hairy sea cucumber *Sclerodactyla briareus*"
- Project 4: "Annotation of the draft genome of the hairy sea cucumber *Ophioderma brevispinum*"
- Primary advisor: Professor Daniel Janies

2020–2021 **College of Computing and Informatics**, *University of North Carolina at Charlotte*, Charlotte–NC, USA
- Level: Master's (project 1) and undergraduate (project 2)
- No. of students: 2
- Project 1: "The fundamental evolution of SARS-CoV-2 and other coronaviruses"
- Project 2: "Genomic epidemiology of SARS-CoV-2"
- Primary advisor: Professor Daniel Janies

2019–2020 **Interunites Graduate Program in Bioinformatics**, *University of São Paulo*, São Paulo–SP, Brazil
- Level: Master's
- No. of students: 1
- Project: "The assembly of the genome of the eastern spadefoot toad, *Scaphiopus holbrookii*"
- Primary advisor: Professor Taran Grant

— **College of Computing and Informatics**, *University of North Carolina at Charlotte*, Charlotte–NC, USA
- Level: Master's
- No. of students: 2
- Project 1: "FLAVi-Web: a web annotator for viral genomes of *Flaviviridae*"
- Project 2: "Assembly and annotation of repetitive DNA in the genome of the golden poison dart frog, *Phylobates terribilis*"
- Primary advisor: Professor Daniel Janies

— **The University Honors Program**, *University of North Carolina at Charlotte*, Charlotte–NC, USA
- Level: Undergraduate
- No. of students: 1
- Project: "Repetitive DNA and tissue regeneration in echinoderms"
- Primary advisor: Professor Daniel Janies

Teaching Interests

I am currently the instructor for **Programming II** (BINF-6112 and BINF-8112, combined) and **Intro to Computer Science II** (ITSC1213-001). I have also served as the host of the Larry Mays **Seminar** Series in Bioinformatics (BINF-6600 and BINF-8600, combined) and as the instructor of **Machine Learning** for Bioinformatics (BINF-6210 and BINF-8210, combined). There are a number of courses that interest me in the Department's Master's and Doctorate programs. I am particularly motivated and qualified to teach technical and methodological courses on **biostatistics**, **database systems**, **big data analyses**, and **health informatics**. I am also interested in courses such as **environmental health**, **epidemiology of infectious diseases**.

I am interested in proposing new workshops and courses on **One Health**, **genomic epidemiology**, and **phylogenetics as applied to infectious diseases**.

Training and Certifications

2022–2026 **Research compliance training (CITI Program)**, *UNC Charlotte*, Charlotte–NC, USA
- Responsible Conduct of Research, Biomedical Sciences and Administrative (2023)
- Biosafety: principal investigators and laboratory personnel (2022), NIH Guidelines for recombinant and synthetic nucleic acid molecules (2023), animal biosafety (2023), and shipping and transport of regulated biological materials (2026)
- Human subjects research: data or specimens only (2025) and biomedical research (2026)
- Research security (2026) and export controls (2026)

2024 **QPR (Question, Persuade, Refer) suicide prevention training**, *UNC Charlotte*, Charlotte–NC, USA
- Date: June 26, 2024

2024 **Career Consultant Training Program (now Career Collaborator)**, *UNC Charlotte, University Career Center*, Charlotte–NC, USA
- Completed January 30, 2024; Career Collaborator badge, August 2024

2023 **Green Dot violence prevention instructor certification**, *Alteristic, at UNC Charlotte*, Charlotte–NC, USA
- Instructor training: February 14–17, 2023

2023 **Charlotte Launch (Spring 2023 cohort)**, *UNC Charlotte*, Charlotte–NC, USA
- Customer discovery program affiliated with NSF I-Corps; certificate, May 2023

2023 **Charlotte Launch 2.0**, *UNC Charlotte*, Charlotte–NC, USA
- Customer discovery program for the mutable collagenous tissue project, with Reyhaneh Nouri and Vladimir Mashanov
- Final presentation: August 2023

2011 **The Willi Hennig Society 12th International Workshop in Phylogenetic Methods**, *Instituto de Ecología*, Xalapa–Veracruz, Mexico
- Duration: 40h
- Support: The Willi Hennig Society

2010 **Theory and Practice in Phylogenetic Reconstruction**, *Institute of Biosciences, University of São Paulo*, São Paulo–SP, Brazil
• Duration: 80h

2009 **Foundations of Systematics and Biogeography**, *Institute of Biosciences, University of São Paulo*, São Paulo–SP, Brazil
• Duration: 80h

2006 **Systematics**, *Universidade Estadual Paulista "Júlio de Mesquita Filho"*, São Vicente–SP, Brazil
• Duration: 40h

Editorial Roles

2026–current **Editorial Board Member**: Cladistics (Wiley), the Willi Hennig Society's journal, where I handle manuscripts as an editor

2024–current **Editorial Board Member**: BMC Genomic Data (Springer Nature)

2026–current **Topic Editor**: Frontiers in Bioinformatics, Research Topic "Volume II: Methods, Tools and Algorithms in Evolutionary Bioinformatics" (with J. M. Craig and S. Miura)

2023 **Topic Editor**: Frontiers in Microbiology, Research Topic "Phylogenetics in the One Health Context" (with L. Jiménez-Ferbans and K. Ocaña). Editorial: DOI 10.3389/fmicb.2023.1307616

2023–current **Guest Associate Editor**: Frontiers in Microbiology, section Evolutionary and Genomic Microbiology

2021–current **Review Editor**: Frontiers in Virology, section Bioinformatic and Predictive Virology

— **Review Editor**: Frontiers in Microbiology, section Virology

— **Review Editor**: Frontiers in Bioinformatics, section Evolutionary Bioinformatics

In October 2026, my Loop profile listed 30 editorial contributions to Frontiers journals: 2 edited Research Topics, 1 edited publication, and 12 reviewed publications. See loop.frontiersin.org/people/1352271/editorial.

Service

2026–2028 **Council Member**: The Willi Hennig Society

2026 **Organizer**: Sarstedt manufacturing plant tour for Biology, Bioinformatics, and Biomedical students (student sign-up, selection criteria, and outreach)

2026–current **Regular Faculty**: EVOLVE, a weekly evolution seminar and journal club at UNC Charlotte

2025–current **Committee Member**: Larry Mays Seminars in Bioinformatics, Department of Bioinformatics and Genomics

2025–2026 **President**: Latinx/Hispanic Staff and Faculty Caucus (LHFS), UNC Charlotte

2025 **Panelist**: "Preparing for the Faculty Interview," Postdoctoral Readiness to Enter the Professorship Program (PREPP), UNC System

2024–current **Co-Chair**: Ad Hoc Review Committee on Faculty Council Standing Committee Structure and Governance Documents, UNC Charlotte (since November 2024)

2024–current **Secretary of the Faculty**: UNC Charlotte Faculty Council and Faculty Executive Committee (since 2024; re-elected in April 2025 for a term ending May 2027)

2024–current **Committee Member**: Ph.D. Steering Committee, Department of Bioinformatics and Genomics (since July 2024)

2024–current **Review Panelist**: UNC Charlotte internal research grants and funding proposals

2024–2025 **Communications Officer**: Latinx/Hispanic Staff and Faculty Caucus (LHFS), UNC Charlotte

2023–2025 **Faculty Council Representative**: Department of Bioinformatics and Genomics (from August 2023)

2023, 2025 **Host**: Larry Mays Seminars in Bioinformatics (BINF 6600/8600), Fall 2023 and Spring 2025

2023–current **Green Dot Trainer**: violence prevention program, UNC Charlotte (since February 2023), with four trainings for 97 participants in 2023 to 2024

Reviews

Up until 2025, I had completed 67 reviews in 28 academic journals. In 2026, through October, I reviewed four more manuscripts (for Cladistics, Mitochondrial DNA Part B, the Journal of Oceanology and Limnology, and a Frontiers journal) and handled ten Cladistics manuscripts as an editor.

13	Frontiers in Microbiology	2	Viruses
5	Cladistics	1	Advanced Science
5	PLoS One	1	Anais da Academia Brasileira de Ciencias
4	BMC Medical Genomics	1	BMC Evolutionary Biology
4	Infectious Disease Reports	1	Clinical and Translational Medicine
3	BMC Genomics	1	Ecology and Evolution
3	Molecular Biology and Evolution	1	EJIHPE
3	Qeios	1	F1000
2	Frontiers in Bioinformatics	1	Foundations of Science
2	Hydrobiologia	1	Frontiers in Virology
2	Mitochondrial DNA Part A	1	Infection, Genetics and Evolution
2	Molecular and Cellular Biochemistry	1	Life
2	Pathogens	1	Mitochondrial DNA Part B
2	PeerJ	1	Wellcome Open Research

For more details, see my Publons profile at publons.com/researcher/665479/denis-jacob-machado (note that Publons does not necessarily show all my reviews).

Skills

Languages

Portuguese Native

English Fluent

Spanish Proficient

Hebrew Beginner

Latin Beginner

Skills in Computational Biology

Concepts used in my current and past projects include high-performance computing, high-throughput nucleotide sequencing, machine learning, protein-protein interaction, protein structure, antibodies, gene expression analyses, metagenomics, Python pipelines, R Shiny web apps, and cloud-based applications (Microsoft Azure and AWS).

Languages `Bash`, `R` (including Shiny applications), `Python`, `Perl`, SQL, `JavaScript`, `Java`, `PHP`, and `Haskell`

Tools `AlphaFold2`, `RoseTTAFold`, `HADDOCK`, `Tableau`, Flask, `Git`, `Jupyter`, and LaTeX

Cluster Management

I have built and managed computer clusters designed for the analyses of biological data from multi-omics research.

2013–2022 **ACE**, *University of São Paulo*, São Paulo – SP, Brazil
- *Ace* is a FAPESP-funded SGI cluster housed in the Museum of Zoology
- Homepage: www.ib.usp.br/gran

2009–2013 **Abacus**, *University of São Paulo*, São Paulo – SP, Brazil
- *Abacus* is a FAPESP-funded Beowulf cluster housed in the Institute of Biosciences
- Homepage: www.ib.usp.br/hpc

Software Development

Most of my published software are available under an open-source license at:

GitLab gitlab.com/machadodj
GitHub github.com/machadodj
Web phyloinformatics.com

Péva Phylogenetics platform and successor to YBYRÁ: parsimony tree search, consensus, tree distances, wildcard terminals, sensitivity plots, character categorization, and resampling. gitlab.com/phyloinformatics/peva-public

RECOSIM Simulates and detects recombination in aligned sequences. gitlab.com/phyloinformatics/recosim

CladesRF Measures the association between terminal metadata and the clades of a tree with random forests. gitlab.com/phyloinformatics/cladesrf

CladesGBM Asks the same question as CladesRF with LightGBM and a perturbation test. gitlab.com/phyloinformatics/cladesgbm

FLAVi Gene prediction and annotation for *Flaviviridae* genomes (gitlab.com/MachadoDJ/FLAVi). FLAVi-CoV adapts it to SARS-CoV-2 and related viruses (gitlab.com/MachadoDJ/flavi-cov).

CMA Gene prediction and annotation for cestode mitogenomes. gitlab.com/MachadoDJ/cma

AWA Tests the completeness of circular DNA assemblies from short reads. gitlab.com/MachadoDJ/awa

YBYRÁ Tools for tree distances, clade sensitivity, synapomorphy categorization, and wildcard taxa. Archived in July 2026, with its functions now in Péva. gitlab.com/MachadoDJ/ybyra

Online Research Profile

LinkedIn linkedin.com/in/machadodj
ORCID 0000-0001-9858-4515
ResearcherID I-1452-2015
Publons publons.com/researcher/665479
Scopus https://www.scopus.com/authid/detail.uri?authorId=55958393200
G Scholar scholar.google.com/citations?user=TK6Ms80AAAAJ
Loop loop.frontiersin.org/people/1352271

Immigration & Citizenship

U.S.A.	Resident alien (Green Card)
Brazil	Citizen (native)